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Voltammetric electronic tongue for the discrimination of antibiotic mixtures in tap water

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ABSTRACT

The identification and quantification of antibiotics in different types of water is of growing interest due to the resistance their presence can induce in many bacterial species. Although the electrochemical analysis of electroactive antibiotics via voltammetric techniques is rapid, simple and cost-effective, their simultaneous detection is often challenging due to their overlapping oxidation peaks. To address this issue, we have developed a voltammetric electronic tongue for the discrimination and quantification of antibiotic mixtures in tap water. For that purpose, three different screen-printed carbon electrodes were modified with "green"-synthesized metallic nanoparticles (NPs) of copper and silver, and commercial titanium dioxide NPs. The electrochemical behavior of the electrodes against the detection of six antibiotics was first investigated by cyclic voltammetry, proving the catalytic effect of the selected NPs. Next, principal component analysis and artificial neural networks (ANNs) were used to achieve the accurate discrimination and quantification of three selected antibiotics, cefepime, daptomycin, and vancomycin, at the mg/L level with a normalized root mean square error (NRMSE) of 0.055 for the test subset. Finally, after validation of the ANN model, the analysis of tap water samples spiked with the three selected antibiotics was carried out to test its applicability for water quality monitoring, showing a good agreement with the expected values (average recovery of 95 %, and NRMSE of 0.034).

1. Introduction

The overuse and release of antibiotics into water bodies exacerbate antibiotic resistance [1–3], considered by the World Health Organization to be an urgent global health issue as pointed out in a report of the Organization for Economic Co-operation and Development. This crisis is expected to cause more deaths worldwide than cancer by 2050 [4]. Since there is no regulation regarding the release of pharmaceuticals into municipal wastewater, a variety of active pharmaceutical ingredients is found in sediments, surface water, groundwater, and drinking water [5]. This is why in recent decades, research has focused on delivering new methods for the rapid detection and identification of antibiotics and other drugs [6–14].

Conventional approaches for the determination of antibiotics are mainly based on liquid chromatography as the gold standard method, especially when coupled with mass spectrometry detection [5,15,16]. However, despite its high performance, this method is rather expensive (with high economic disbursement for acquisition and maintenance), non-suitable for on-site analysis, requires experienced and trained personnel, and uses large volumes of organic solvents [5,17]. As an alternative, electrochemical sensors have risen as a rapidly developing technology of portable devices, offering low-cost and compact instrumentation [18]. Indeed, the use of electrochemical sensors has been broadly reported to facilitate the rapid and sensitive detection of trace levels of electroactive antibiotics in water, mostly based on measuring the current generated from the oxidation of those antibiotics when the

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right potential is applied [19]. Nonetheless, the selectivity of these sensors falls short when quantifying simultaneously various antibiotics in a sample, especially when those antibiotics are oxidized at similar oxidation potentials. To overcome selectivity issues, various strategies have been reported, most of them based on the use of customized modifications of the transducer with different catalysts such as carbon allotropes, metallic nanoparticles (NPs) or (bio)recognition elements [20,21]. By incorporating a catalyst within the sensor design, highly selective and sensitive devices that are fully operational in real conditions can be delivered. Salihu *et al.* developed a sensitive electrochemical sensor for penicillin G in bovine milk samples using a screen-printed carbon electrode (SPCE) modified with nickel NPs [22]. Similarly, Muhammad *et al.* developed a sensitive electrochemical sensor for the determination of thiamphenicol residue in milk using a SPCE modified with carbon nanotubes and gold NPs [23]. Precisely, advances in materials science and nanotechnology have underpinned the evolution of these modified sensors and spurred their potential [24,25]. The use of electrodes modified with NPs has arisen over the last years as a promising approach for the development of electrochemical sensors with enhanced analytical performance, due to the unique properties of NPs in comparison to the respective bulk materials [26–29]. NPs have unique physical and chemical properties, derived from their size and high surface area [30]. By selecting NPs size, shape and structure, interesting catalytic properties can be achieved [31,32]. In this direction, many metallic NPs have been evaluated as modifiers for voltammetric electrodes, gold, silver, and platinum NPs likely being the more common ones. A trend that could be related to their typical use as bulk electrode materials. However, similar behavior has been reported for other less common metals such as nickel or copper [27]. In addition, over the last years, with the increase in environmental awareness, the focus has shifted towards developing "green"-synthesized materials to modify sensors [34,35]. Such approach aims to avoid the production of unwanted or harmful by-products through the build-up of reliable, sustainable, and eco-friendly synthesis procedures [34]. For example, in the case of metallic NPs, by producing them from plant extracts instead of industrial chemical agents to reduce metal ions [36].

The main advantages of NPs in regard to electroanalysis include their high effective surface area, enhanced mass transport, catalysis and, as a result, improved signal-to-noise ratio [28,33]. The combination of the enhanced mass transport and catalytic properties can change the apparent peak potential for certain compounds, a situation that is particularly advantageous as it can lead to the separation of the peak potentials of compounds that, otherwise, would overlap, or to different sensitivities between those [28]. Such electrocatalytic effect, shown by sensors modified with certain types of metallic NPs, can significantly improve the discrimination between electroactive species present in a sample when electrochemically detected. As an example, Xue *et al.* reported a nanocomposite consisting of silver NPs and multi-walled carbon nanotubes on black phosphorene that allowed the voltammetric monitoring of uric acid, xanthine, and hypoxanthine simultaneously [40]. Similarly, Yang *et al.* harnessed the electrocatalytic effect of a nanocomposite consisting of ferrocene-modified gold NPs and carbon dots, deposited on a graphene-modified glassy carbon electrode, to simultaneously detect ascorbic acid, dopamine, uric acid, and acetaminophen via electrochemical means [41]. Even though advances have been reported in the simultaneous electrochemical detection of species that feature similar oxidation potentials, there are still some key challenges to face when quantifying electroactive antibiotics in a complex matrix.

In this direction, electronic tongues (ETs) have shown great potential in addressing the challenges involved in the simultaneous quantification of various electroactive species in a complex matrix [42–45]. ETs are multi-sensor systems consisting of an array of sensors showing cross-reactivity that harness chemometrics to overcome the convoluted nature of their responses, allowing to obtain comprehensive information about samples by simultaneously detecting a large spectrum of compounds. Thus, ETs are particularly suited for the analysis of complex mixtures.

Herein we report the development of a voltammetric ET for the analysis of antibiotic mixtures in water samples. The research proposed aims to offer a cost-effective, rapid, and accurate solution for the simultaneous detection and quantification of antibiotic mixtures in tap water samples, overcoming the limitations of the analytical methods traditionally used. The sensor array consisted of three different electrodes modified with commercial TiO_2 NPs, and "green"-synthesized Ag and Cu NPs, respectively. "Green"-synthesized Ag and Cu NPs were chosen due to their attractive properties [26,37,38], while also offering a much lower cost in comparison to *e.g.,* gold NPs. TiO_2 NPs were selected due to their well-known use in heterogeneous catalysis (including the degradation of pollutants) [39]. Cyclic voltammetry (CV) was first used for the individual characterization of the sensors' responses toward several antibiotics, including cefepime (Cef), daptomycin (Dap), and vancomycin (Van). Next, an Artificial Neural Network (ANN) model was built to achieve the simultaneous determination of ternary mixtures of the aforementioned antibiotics in tap water samples, aiming to improve the accuracy and reliability of detecting multiple antibiotics, making this approach highly suitable for real-world applications in water quality monitoring.

2. Materials and methods

2.1. Reagents and apparatus

Silver nitrate ($AgNO_3$, purity$\geq$ 99.0 %), and copper sulfate ($CuSO_4$, purity $\geq$ 99.0 %), and phosphate buffer saline tablets (PBS) were purchased from Merck (Germany). PBS tables were reconstituted in ultrapure water (18.20 MΩ·cm), to prepare 10 mM phosphate buffer, with 2.7 mM potassium chloride and 137 mM sodium chloride, and pH adjusted to 7.4 (PBS). Aztreonam (Bristol Myers Squibb), azithromycin (Altan Pharmaceuticals), cefepime (Cef; Fresenius Kabi España), daptomycin (Dap; Merck Sharp & Dohme), meropenem (Sun Pharmaceutical industries, Europe B.V) and vancomycin (Van; Reig Jofre) were the antibiotics considered in this study. Titanium dioxide NPs (particle size 21 nm) with anastase structure were purchased from Sigma-Aldrich (St. Louis, MO, USA), while copper NPs (particle size 20 nm) and silver NPs (particle size 45 nm) were synthesized following the protocols previously reported by the authors [46,47]. These NPs were used to modify the working electrode within each SPCE.

The electrochemical measurements were carried out using a DropSens μStat 8000 controlled with DropView software (v 2.21). Voltammetric measurements were conducted using three SPCEs (each modified with one type of NPs) consisting of a carbon working electrode (3 mm diameter), a pseudo-reference electrode made of Ag/AgCl, and a carbon counter electrode.

2.2. Samples preparation

Stock solutions of each of the antibiotics were prepared in PBS, from which the different standards were obtained. Firstly, to check the voltammetric behavior of each modified electrode towards each antibiotic, stock solutions of aztreonam (108 mg/L), azithromycin (186 mg/L), Cef (120 mg/L), Dap (405 mg/L), meropenem (95.5 mg/L) and Van (362 mg/L) were used. Next, individual calibration curves were built for different antibiotics to characterize the analytical response of the sensors in terms of sensitivity, linear range, limit of detection (LOD), reproducibility, etc. For this purpose, solutions of increasing concentrations of each antibiotic were prepared in PBS and measured under the conditions described in Section 2.3.

Finally, for the simultaneous determination of Cef, Dap, and Van, a set of samples made up of a mix of the three antibiotics was prepared. The samples were divided into two subsets: a training subset formed by

27 samples based on a tilted 3^3 factorial design used to fit the model, and a testing subset formed by 20 samples distributed along the experimental domain defined by the former and used to evaluate the model performance. Moreover, to test the applicability of the developed ET for water quality monitoring, 8 tap water samples spiked with various concentrations of the three antibiotics were also prepared.

2.3. Sensors' preparation and electrochemical measurements

Stock solutions of NPs were firstly sonicated for 15 min to prevent aggregation. Then, various SPCEs were modified with one of the three selected metallic NPs by drop-casting 2 μL of a 0.9 mM Ag (0.194 μg), 5 mM Cu (0.635 μg), or 10 mM TiO_2 (1.59 μg) NPs solution onto the working area of the SPCEs, and then cured at 80 °C for 5 min to facilitate the adhesion and stability of the NPs on the electrode surface. Afterwards, the electrodes were rinsed with Milli-Q water and stored at room temperature until used.

Voltammetric fingerprints of the different antibiotics were obtained by CV by sweeping the potential between 0 V and 1 V *vs.* Ag/AgCl at a scan rate of 100 mV/s. The same conditions were employed for the individual characterization, as well as the analysis of mixtures, without any other precaution than performing a blank measurement in 10 mM PBS (pH 7.4) after each measurement to confirm no passivation of the electrodes.

Additionally, amperometric measurements at a fixed potential were taken to check the stability of the electrodes after successive measurements. Concretely, current intensities for the oxidation of 120, 405, and 362 mg/L stock solutions of Cef, Dap, and Van at 0.67 V, 0.72 V, and 0.58 V, respectively, were measured.

2.4. Data analysis

Data processing was done in Matlab R2020b (Mathworks, Natick, MA, USA) by specific routines developed by the authors making use of its Statistics and Machine Learning toolboxes. For the qualitative analysis, obtained voltammetric responses of the stock solutions were submitted to principal component analysis (PCA), while for the quantitative analysis, an artificial neural network (ANN) model was built using discrete wavelet transform (DWT) as feature extraction tool. The use of DWT allows to compress the voltammetric data without significant loss of information by preserving only the low-frequency coefficients and discarding the rest [48]. The use of a data reduction step has demonstrated it can improve the model performance and generalization ability, while also providing further advantages during the modeling step. Similarly, ANNs were chosen given their high modeling performance and suitability to model both linear and non-linear relationships [48, 49].

3. Results and discussion

3.1. Voltammetric responses

Upon modification of the different SPCEs, the first step was assessing the catalytic properties of the selected NPs and optimizing the amount of NPs deposited. To this aim, 108 mg/L aztreonam, 186 mg/L azithromycin, 120 mg/L Cef, 405 mg/L Dap, 95.5 mg/L meropenem, and 362 mg/L Van solutions were measured via CV using each modified sensor. The obtained voltammograms are shown in Fig. 1, where a clear catalytic effect can be observed, in regard to both the current and potential of the oxidation peaks. For instance, if Dap is taken as an example, it can be seen how the oxidation peak shifts to lower potentials following the order Ag, Cu and TiO_2 NPs, respectively. More importantly, there is not only a shift in the potential, but the oxidation peak also becomes more prominent and clearer. A similar behavior is observed for Cef, where the shift of the peak is even higher than 100 mV. Oppositely, despite the good catalytic properties shown, it should be noted that the sensors modified with TiO_2 NPs show larger capacitive currents among all NP-modified sensors, which is attributed to the TiO_2's smaller electrical conductivity in comparison to Cu or Ag.

Overall, the voltammograms depicted in Fig. 1 confirm the benefits derived from the usage of the different NPs selected. The potential of NPs to improve the electrochemical detection of each antibiotic relies on their ability to boost the electrochemical signal measured upon interacting with each antibiotic. As a result of the interactions between metal NPs and the functional groups on each antibiotic tested, the electrochemical signal measured is amplified, and the oxidation potential shifted, contributing to enhance the electrochemical performance of the modified sensors. Cef is a cephalosporin with several functional groups, including amide and carboxyl groups, which can interact with metal NPs via adsorption or complex formation [50]. The lipopeptide Dap has a cyclic peptide structure with several carboxyl and hydroxyl groups that can easily interact with metal NPs [51]. Indeed, Dap's peptide bonds can form coordination complexes with NPs. Finally, the interaction between the glycopeptide Van and metal NPs has been reported to occur through the hydroxyl and amino groups of the former [52].

It should be noted that, as described in the literature [51,53], peaks can also be shifted if measurements are carried out at different pH. Here, PBS with neutral pH was selected due to the fact that the pH of both drinking and wastewater is within the pH range 6–8. Consequently, conducting the experiments at pH 7.4 ensured that our findings are directly applicable to the conditions expected for the analysis of both drinking and wastewater samples.

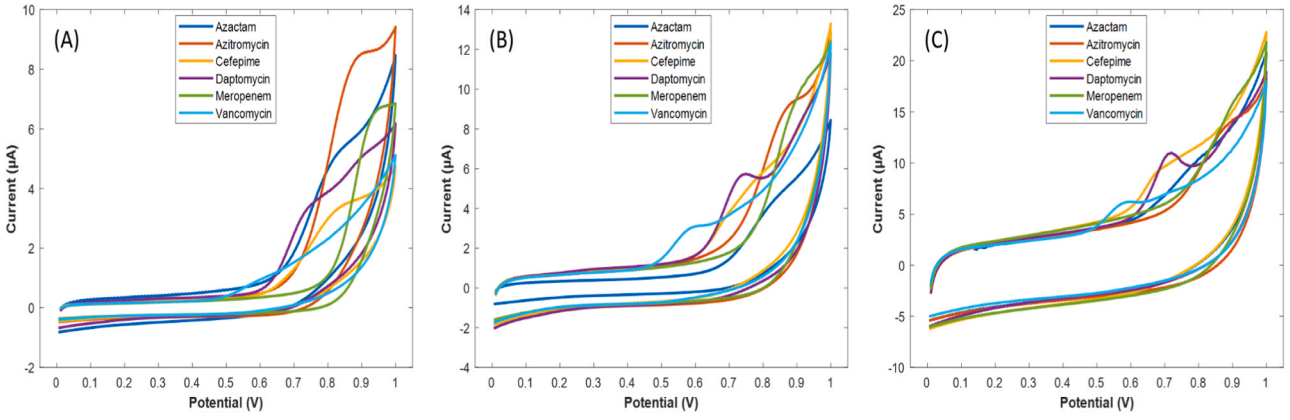


Fig. 1. Cyclic voltammograms recorded at 100 mV/s when measuring equimolar antibiotic solutions (corresponding to the following mg/L concentrations: 108 mg/L aztreonam, 186.5 mg/L azithromycin, 120 mg/L Cef, 405 mg/L Dap, 95.5 mg/L meropenem, and 362 mg/L Van) prepared in 10 mM PBS (pH 7.4), using SPCEs modified with (A) Ag NPs, (B) Cu NPs and (C) TiO_2 NPs.

3.2. Qualitative analysis

Next, to assess the potential of the developed sensors to successfully discriminate among antibiotics, the obtained responses were submitted to PCA. PCA represents the use of a rapid visualization tool that allows to assess samples (dis)similarities by reducing the dimensionality of the original dataset into a few principal components (PCs) with minimal information loss, that contain most of the original variance of the data.

To this aim, three replicate voltammetric measurements of 250 μM antibiotic solutions (*i.e.,* 108 mg/L aztreonam, 186 mg/L azithromycin, 120 mg/L Cef, 405 mg/L Dap, 95.5 mg/L meropenem, and 362 mg/L Van) were carried out. The obtained scores plot is shown in Fig. 2, where clear clusters can be observed for each of the antibiotics considered. First of all, it can be seen how *ca.* 72 % of the variance present in the original data (1001 × 3 points/sample) is now summarized with only the first 2 PCs. More importantly, some clusters can be observed, which correspond to the replicate measurements of the different antibiotics, while measurements corresponding to the different antibiotics appear far away from each other. To statistically evaluate the relevance of the clusters observed, as PCA is an unsupervised method that lacks a learning rule that allows the classification of different samples into groups, the 95 % confidence intervals for the different samples were calculated and plotted [54]. As it could be expected, those are narrow and do clearly not overlap. These results confirm the capability of the proposed sensor array to extract a characteristic fingerprint of each antibiotic that allows its discrimination.

Furthermore, the Silhouette index was calculated to simultaneously assess the cross-response and repeatability/reproducibility of the sensors in a single step, harnessing the results from the PCA plot. The key hypothesis that interconnects both is that the cohesion of the clusters can be interpreted as a measure of the sensors' repeatability/reproducibility (depending on whether replicates of each type of sensor are used or not), while their separation can be taken as a measure of the sensor capability to discriminate the targeted compounds. In other words, well-defined and separated clusters in the scores plot, might indicate sensors showing good repeatability/reproducibility and selectivity. Thus, by using a clustering metric such as the Silhouette index an overall metric of the performance of the ET can be provided. Silhouette corresponds to the chi-square distance between *j*-dimensional points, and its value (which can range from −1 to +1) provides a measure of how similar a point is to points in its own cluster, when compared to points in other clusters [55]. In this case, the calculated Silhouette index value was 0.993, a high value close to its maximum.

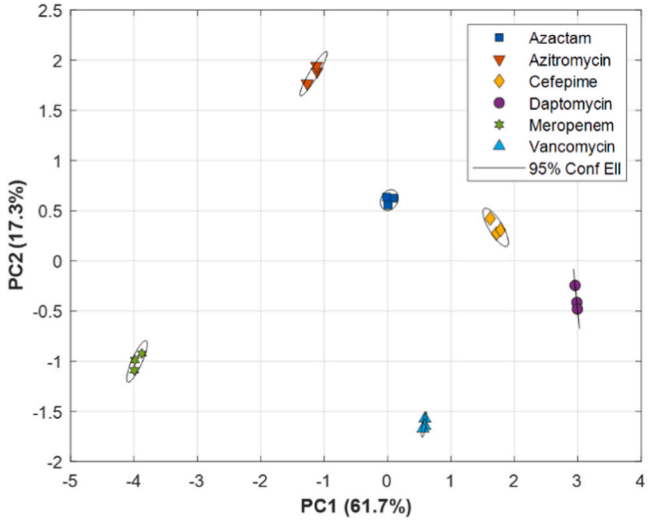


Fig. 2. PCA scores plot of the voltammetric responses of the three sensors towards different antibiotics. The ellipses plotted correspond to the 95 % confidence limits for each of the clusters.

However, despite the good discrimination obtained when measuring stock solutions of each antibiotic, the truth is that when mixtures of antibiotics are to be analyzed, the complexity of the voltammetric signals increases as cross-effects are observed. For example, if Dap or Van are to be analyzed with the TiO_2-modified sensor, an easily distinguishable signal for each compound is obtained (Fig. 3). Nevertheless, when an equimolar mixture of those is analyzed (Fig. 3b), Van peak is no longer observed, but a broader shoulder is obtained prior to the actual Dap peak. This situation gets even more complicated if not only two antibiotics are mixed, but three or more (as shown in Section 3.4).

Thus, with the aim of proving the potential of the approach to overcome the signal overlapping and undesired effects, and to achieve the individual quantification of antibiotics in mixtures of them, the antibiotics whose clusters appeared closer in the PCA scores plot were selected as the study case (*i.e.,* Cef, Dap and Van).

3.3. Analytical characterization of the sensors' response

The next step, before moving on to the simultaneous quantitative determination of different antibiotics, was the analytical characterization of the responses of the different electrodes. On the one side, when developing any sensor device, it is important to assess its sensitivity, LOD, working range, etc. On the other side, despite the great advantages that the data processing stage may offer (*e.g.,* allowing to compensate non-linearity, drifts or overlapping in the responses, or offset matrix effects from the sample itself), the stability of the sensors' response together with their varied cross-sensitivity is critical. Therefore, when developing an ET application, it is of utmost importance to ensure the sensors show cross-response towards the different analytes, as well as the reproducibility of the measurements to minimize the effect of the variability in the response of the large number of measurements required to build the chemometric model.

Consequently, individual calibration curves for each antibiotic were constructed by measuring stock solutions of increasing concentrations (Fig. 4) under the voltammetric conditions described in Section 2.3. Next, corresponding peak's heights were plotted against the concentrations to get the fitted regression equations, from which the sensitivity, linearity, and LOD were calculated. The latter was estimated as the concentration corresponding to three times the standard deviation of the blank divided by the slope obtained from the linear fitting. The results are summarized in Table 1, where the high values of the coefficient of determination (R^2) confirm the good linearity at all modified electrodes with respect to all antibiotics. More importantly, different sensitivities were obtained among the different electrodes and antibiotics, confirming the cross-responses required to develop the ET approach.

To evaluate the stability and repeatability of the modified sensors, 10 successive measurements of the different antibiotics were conducted. To ensure that no fouling occurred at the electrode surfaces, the sensors were cleaned using PBS and a blank measurement was performed under the same conditions. From those measurements, the relative standard deviation (RSD) was calculated, being *ca.* 2.5 % the overall average value for all the measurements and <4 % for the worst case as shown in Table S1 (Supporting Information). These results confirm the good stability of the electrodes and the lack of surface passivation.

3.4. Quantitative determination of antibiotic mixtures

Upon completion of the individual characterization of the sensors' responses, the last step was to attempt the simultaneous determination of the three selected antibiotics. However, despite different voltammetric responses were obtained when those were tested individually, when mixed, their signals overlapped. Therefore, the use of a multivariate regression model that relates the complex voltammetric signals with the individual concentration of each compound was required.

To this aim, a set of samples consisting of mixtures of the three

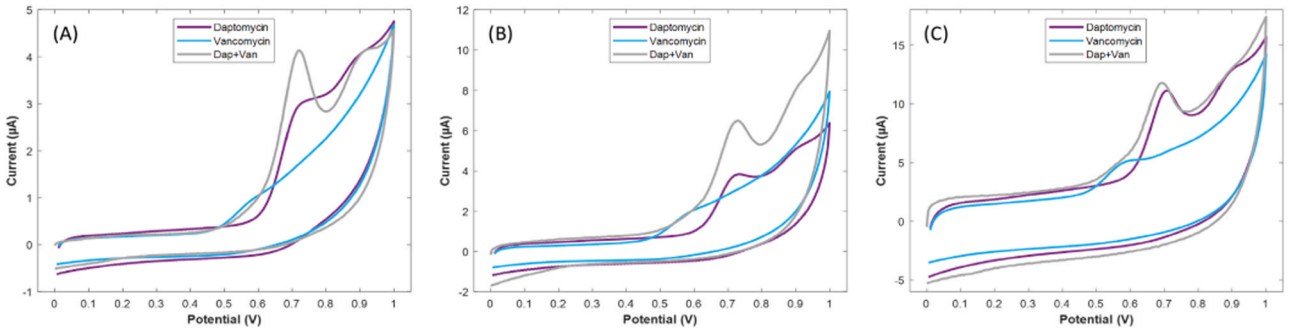


Fig. 3. Cyclic voltammograms recorded at 100 mV/s when measuring 405 mg/L Dap and 362 mg/L Van solutions, prepared in 10 mM PBS (pH 7.4), using SPCEs modified with (A) Ag NPs, (B) Cu NPs and (C) TiO_2 NPs.

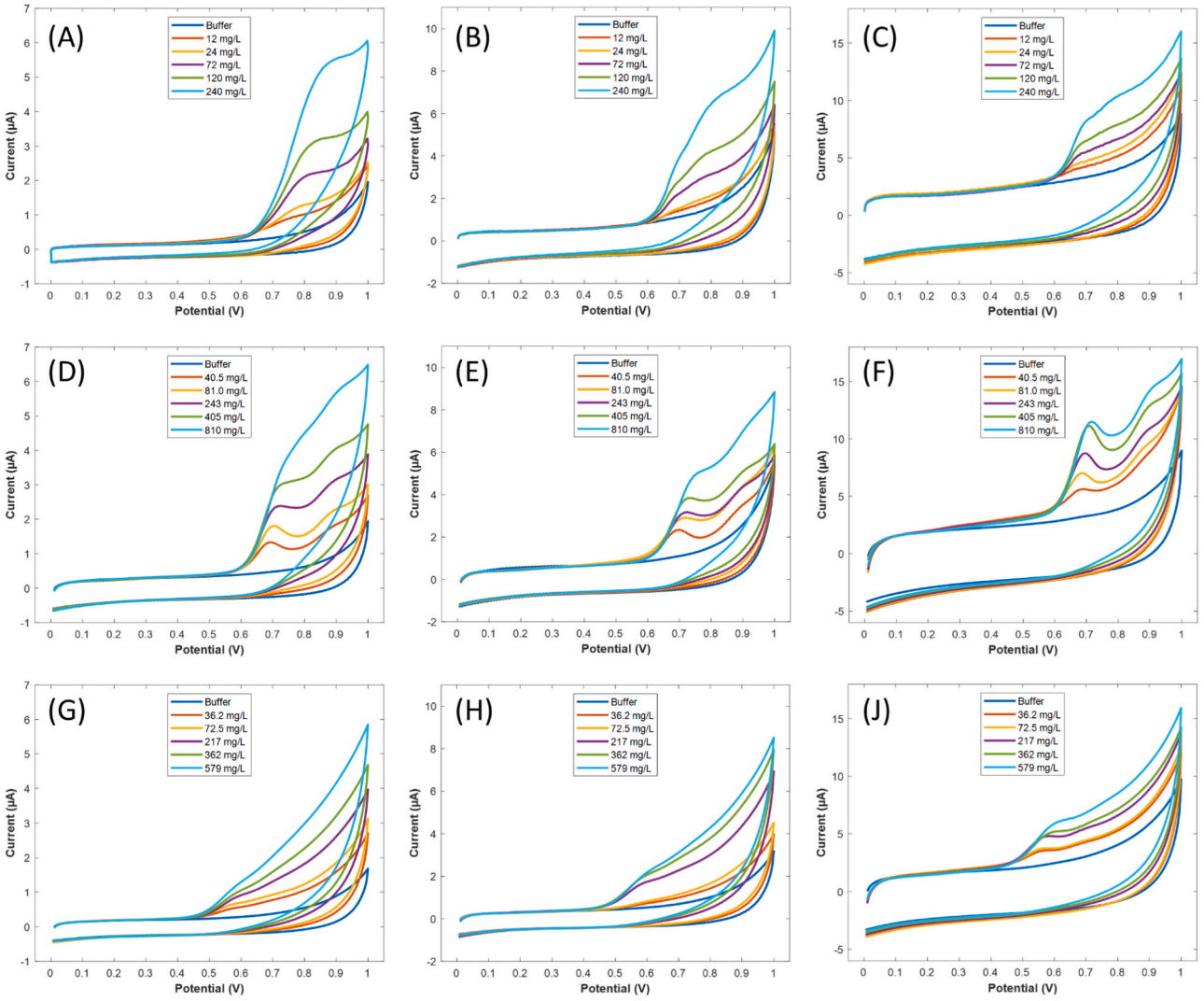


Fig. 4. Cyclic voltammograms recorded at 100 mV/s when measuring solutions of increasing concentrations of Cef (A-C; top row), Dap (D-F; middle row) and Van (G-J; bottom row) in 10 mM PBS (pH 7.4), using the SPCEs modified with AgNPs (A,D,G; left column), CuNPs (B,E,H; middle column) and TiO_2 NPs (C,F,J; left column).

antibiotics was prepared and measured as previously done (Figure S1). Next, the obtained responses were modeled by means of ANNs, which were chosen as the modeling tool given their high performance and ability to model both linear and non-linear responses. Prior to building the model, the voltammetric responses were compressed by means of DWT, and the selected coefficients were fed to the ANN. The model was then fitted using the data of the training subset, while the test subset was used to obtain a more unbiased indicator of its performance. For the selection of the ANN architecture, a systematic study varying the number of neurons in the hidden layer and the transfer functions in both the hidden and the output layers was carried out. Concretely, the number of neurons was changed from 1 to 8, and combinations of *logsig* (log

Table 1
Analytical characterization of the individual response of each sensor towards Cef, Dap and Van.

Antibiotic		TiO_2 NPs-modified SPCE	Green Cu NPs-modified SPCE	Green Ag NPs-modified SPCE
Cef	Sensitivity ($nA/mg{\cdot}L^{-1}$)	4.131	3.582	3.819
	R^2	0.9932	0.9993	0.9954
	LOD (mg/L)	15.58	2.897	8.810
	Oxidation peak (V)	0.6764	0.7752	0.8335
Dap	Sensitivity ($nA/mg{\cdot}L^{-1}$)	7.700	4.894	4.312
	R^2	0.9989	0.9904	0.9885
	LOD (mg/L)	10.56	10.54	12.57
	Oxidation peak (V)	0.6939	0.7008	0.7202
Van	Sensitivity ($nA/mg{\cdot}L^{-1}$)	0.8854	0.5433	0.4441
	R^2	0.9983	0.9997	0.9985
	LOD (mg/L)	5.391	12.62	7.911
	Oxidation peak (V)	0.5746	0.5837	0.5801

sigmoid), purelin (linear) and *satlins* (symmetric saturating linear) transfer functions were tested. Once all the different ANN models were built, their normalized root mean square error (NRMSE) was calculated, and an overall comparison was performed, as shown in Figure S2. Finally, the combination having the lowest NRMSE was selected. In this manner, the final ANN model had 57 neurons in the input layer (corresponding to 19 coefficients x 3 sensors), 5 neurons and *logsig* transfer function in the hidden layer, and 3 neurons (one for each antibiotic) and *satlins* transfer function in the output layer.

To assess the performance of the model, comparison graphs of predicted *vs.* expected concentrations were built (Fig. 5), from which the regression parameters were also estimated to numerically evaluate its goodness. As it can be seen from the plot, a good trend was obtained in all cases, with regression lines close to the ideal line y = x. This is further confirmed by the numerical values, which are not only close to the ideal values (slope equal to 1 and intercept to 0), but within the confidence intervals (CIs). Consequently, the potential of the proposed ET for the qualitative and quantitative analysis of different antibiotics is confirmed.

Finally, after confirming the suitability of the model with the samples from the testing subset, the same approach was applied for the estimation of the concentration of antibiotics spiked in tap water samples with the purpose of testing its applicability for environmental monitoring purposes and validating its performance. Spiked tap water samples were diluted in PBS to maintain the same measuring conditions, without requiring any additional sample pretreatment. Recorded voltammograms were then interpolated into the previously built ANN model, and the corresponding concentrations were obtained. The comparison graphs were also built (Fig. 6) and the regression parameters were calculated.

As it can be seen, good agreement was obtained between the expected and found values, with similar performance obtained for the external test subset and the real samples. Additionally, the recovery values were calculated for each antibiotic and sample, obtaining an overall averaged value of 94.7 %. These results confirm not only the suitability of the ET to overcome the lack of specificity and resolve the overlapped signals, but also to counterbalance undesired effects from the matrix itself.

4. Conclusions

Herein, a voltammetric ET has been reported for the identification and quantification of various antibiotics in tap water. The sensor array consisted of three SPCEs modified with commercial TiO_2 NPs, and

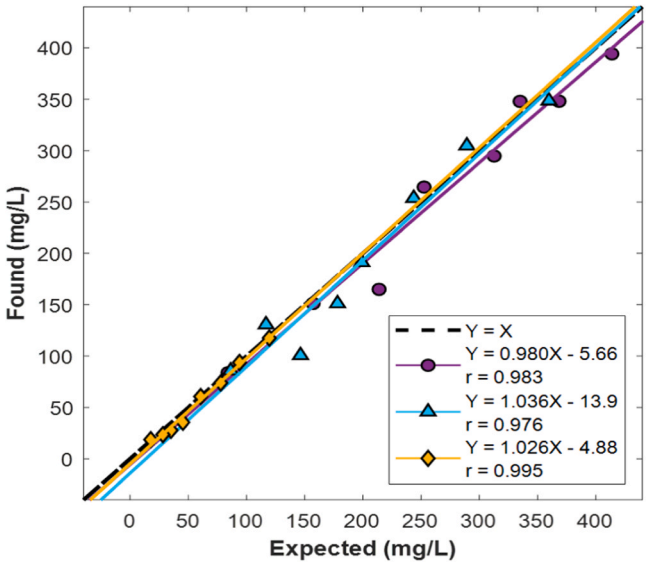


Fig. 6. Applicability of the optimized DWT-ANN model. Comparison graphs of obtained *vs.* expected for tap water samples spiked with Cef (yellow ◆), Dap (purple ●) and Van (cyan ▲). The dashed line corresponds to the ideal comparison line (y = x).

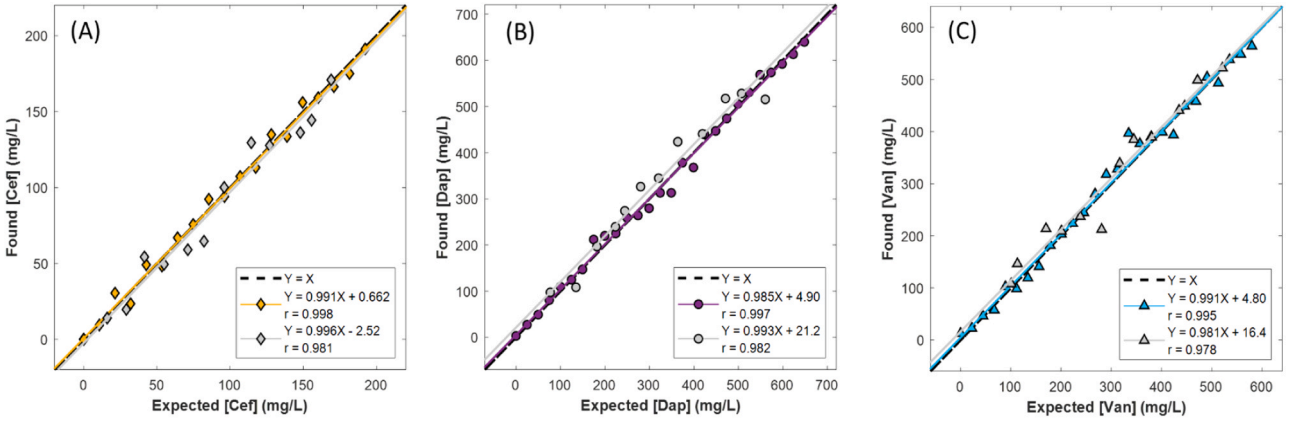


Fig. 5. Modelling ability of the optimized DWT-ANN model. Comparison graphs of obtained *vs.* expected for (A) Cef, (B) Dap and (C) Van, both for the training (colored line and symbols) and testing subsets (grey line and symbols). The dashed line corresponds to the ideal comparison line (y = x).

"green"-synthesized Cu NPs and Ag NPs. The use of different NPs has been shown to be an effective way to generate different electrochemical responses against the different antibiotics, which in combination with pattern recognition methods such as PCA allowed the discrimination of those. Moreover, the quantification of mixtures of antibiotics has been achieved by means of ANNs, which allowed to overcome the overlapping observed in the voltammetric signals.

The main advantages of the proposed approach lie in the combination of the inherent advantages of electrochemical methods, which involve small, cheap, and fast-response portable devices, along with those of chemometrics, which allow building multivariate calibration models to estimate sample properties that would, otherwise, require further sample pre-treatment or the use of more sophisticated methods. By providing a robust tool for the simultaneous detection of multiple electroactive species, this work paves the way towards the monitoring of antibiotic contamination, as a means to ensure water quality monitoring and contribute to global efforts to mitigate its potential impact on the emergence of antibiotic resistance. More importantly, the same approach can be tuned for the analysis of other antibiotics that might be more relevant in other scenarios or even to attempt their determination in biological samples as a monitoring tool in clinical diagnostics. Overall, it can be concluded that the findings reported herein are potentially significant for both environmental and health monitoring, especially when decentralized monitoring is desired. Nevertheless, further validation would be required prior to the application of the developed voltametric ET to health monitoring as no biological samples were tested yet, being beyond the scope of the present study.

CRediT authorship contribution statement

Menyar Ben Jaballah: Writing – original draft, Methodology, Investigation, Formal analysis. **Xavier Cetó:** Writing – original draft, Supervision, Methodology, Investigation, Formal analysis. **Chérif Dridi:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Beatriz Prieto-Simón:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jece.2024.113831.

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